

Natural Resources

Everything we use — the electricity in your home, the gas in a car, the water from your tap — comes from Earth's natural resources. Some refill themselves; others can run out. In this chapter you'll learn which is which, and how the choices we make today shape the planet's future.



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WHAT YOU'LL LEARN

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|--------------------------------|-----------------------------|
| 01 What Are Natural Resources? | 02 Renewable Resources |
| 03 Nonrenewable Resources | 04 Conserving Our Resources |
| 05 A Warming Planet | 06 What We Can Do |



6.1 What Are Natural Resources?

A **natural resource** is anything from the Earth that people use. The water you drink, the air you breathe, the soil that grows your food, the metals in your phone, and the fuels that power cars and homes are all natural resources.

Scientists sort natural resources into two big groups based on one key question: **can it refill itself faster than we use it?**

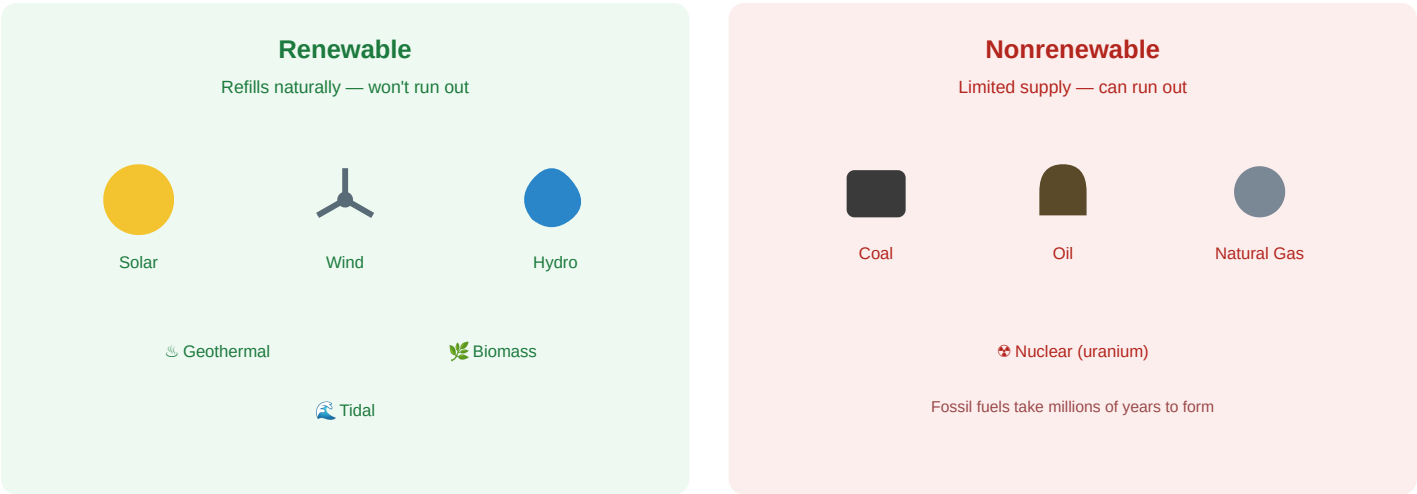


Figure 6.1 — Renewable resources refill naturally and won't run out. Nonrenewable resources exist in limited amounts and can run out.

Type	What It Means	Examples
Renewable	Refills naturally in a short time; won't run out if used wisely	Sunlight, wind, water, geothermal, biomass, tides
Nonrenewable	Exists in a limited amount; takes millions of years to form, so it can run out	Coal, oil, natural gas, uranium (nuclear)

? Quick Check 6.1

1. What is a natural resource?
2. What is the key question that decides if a resource is renewable or nonrenewable?

6.2 Renewable Resources

Renewable resources are nature's refillable energy. As long as the Sun shines, the wind blows, and water keeps flowing, these sources keep coming back. They're sometimes called "clean" or "sustainable" energy because using them usually creates little or no pollution.

Resource	Where the Energy Comes From
Solar	Sunlight captured by solar panels and turned into electricity
Wind	Moving air spins giant turbines to make electricity
Hydro (water)	Falling or flowing water spins turbines, often at dams
Geothermal	Heat from inside the Earth warms buildings and makes electricity
Biomass	Burning plant material or waste to release stored energy
Tidal	The rise and fall of ocean tides turns turbines

★ Why "Sustainable"?

Sustainable means we can keep using it without using it up. Solar and wind are sustainable because the Sun and wind aren't reduced when we use them — tomorrow there will be just as much sunshine and wind as today.

Geothermal: "geo" means Earth and "thermal" means heat — so geothermal energy is literally heat from inside the Earth, the same heat you learned about in the mantle and core.

? Quick Check 6.2

1. Name three renewable energy resources.
2. Why are renewable resources called "sustainable"?

6.3 Nonrenewable Resources

Nonrenewable resources exist in fixed amounts. Once we use them up, they're gone for human lifetimes because they take millions of years to form. Most of the energy the world uses today still comes from these sources.

Fossil Fuels

The most common nonrenewable resources are **fossil fuels**: **coal**, **oil**, and **natural gas**. They formed over millions of years from the buried remains of ancient plants and tiny sea creatures, squeezed and heated deep underground. We burn them to power vehicles, generate electricity, and heat buildings.

Nuclear Energy

Another nonrenewable source is **nuclear energy**, which comes from **uranium**. Nuclear plants split uranium atoms to release huge amounts of energy. Uranium is a mined metal in limited supply, so it's nonrenewable too.

! The Trade-Off

Fossil fuels are powerful and (for now) cheap, which is why we rely on them. But there are two big downsides: they will eventually **run out**, and burning them releases **greenhouse gases** that contribute to a warming planet — which you'll explore in the next section.

? Quick Check 6.3

1. Name the three fossil fuels.
2. Why are fossil fuels considered nonrenewable?
3. What metal is used in nuclear energy?

6.4 Conserving Our Resources

Three of our most precious resources — **water**, **soil**, and **air** — are easy to take for granted, but they can be polluted or used up. **Conservation** means using resources carefully so they last and stay clean for the future.

Resource	The Problem	Ways to Protect It
Water	Fresh water is limited; pollution makes it unsafe	Fix leaks, take shorter showers, keep chemicals out of drains and rivers
Soil	Erosion and over-farming strip away fertile soil	Plant cover crops, plant trees, rotate crops, avoid clearing too much land
Air	Burning fuels releases pollution and greenhouse gases	Use cleaner energy, drive less, reduce factory emissions

★ The Three R's

A simple everyday way to conserve resources: **Reduce** (use less), **Reuse** (use things again instead of throwing them out), and **Recycle** (turn old materials into new products). Each one means we pull fewer new resources from the Earth.

? Quick Check 6.4

1. What does conservation mean?
2. Name one way to protect soil from erosion.
3. What are the "Three R's"?

6.5 A Warming Planet

Scientists who study Earth have measured something important: over the past **century**, the planet's average temperature has slowly but steadily **risen**. Understanding why connects almost everything in this chapter.

Global Temperature Has Risen Over the Past Century

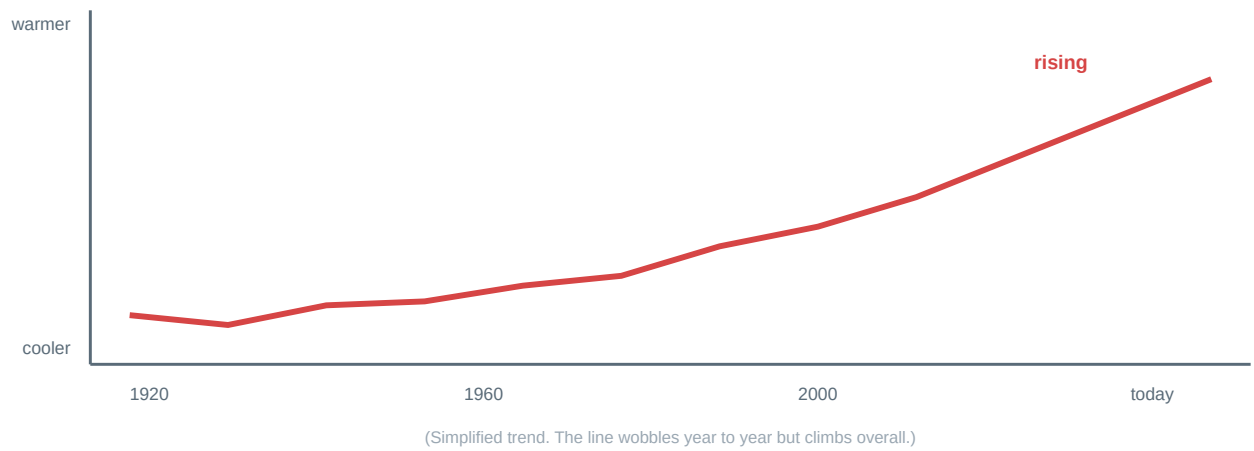


Figure 6.2 — A simplified view of global average temperature over the last 100 years. It bounces up and down year to year, but the long-term trend climbs.

The Connection to Fossil Fuels

Remember from Chapter 4 that greenhouse gases trap heat near Earth's surface. When we burn fossil fuels for energy, we release **extra** greenhouse gases — especially **carbon dioxide** (CO_2) — into the air. More greenhouse gases trap more heat, and the extra heat slowly raises the planet's average temperature.

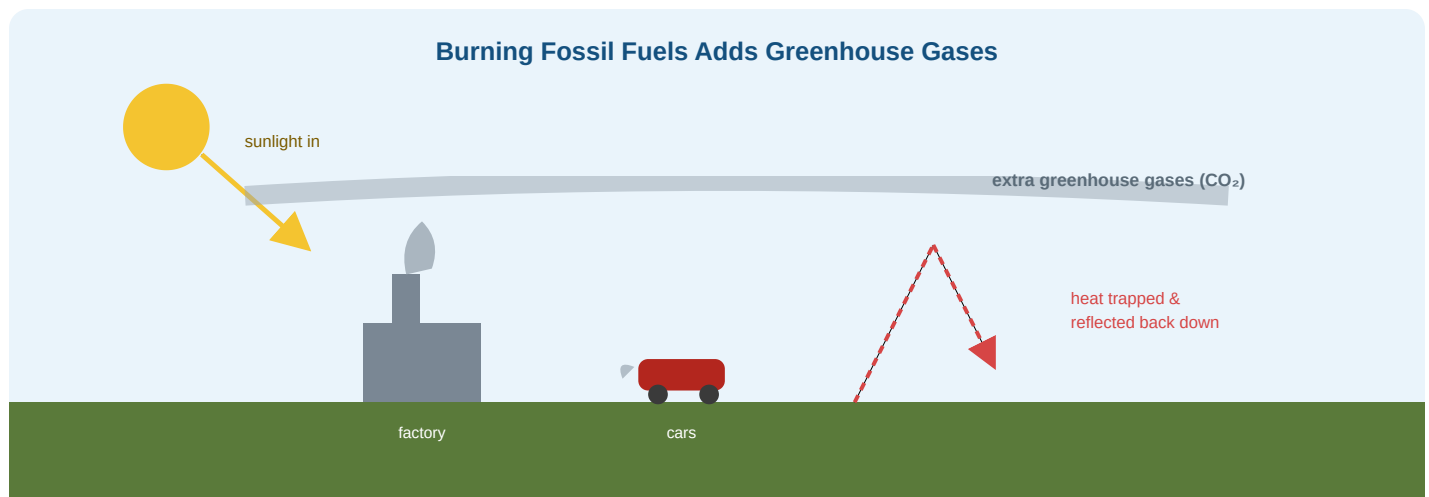


Figure 6.3 — Factories and vehicles burning fossil fuels add extra CO_2 to the air, which traps more heat near the surface.

Scientists look at many lines of evidence — temperature records, melting glaciers, rising sea levels, and the amount of CO_2 in the air — and the data point the same way: the rise in global temperature over the past century lines up closely with the rise in greenhouse gases from human activity.

? Quick Check 6.5

1. What has happened to Earth's average temperature over the past century?
2. Which gas released by burning fossil fuels is a major greenhouse gas?
3. Name one piece of evidence scientists use to study a warming planet.

6.6 What We Can Do

The good news: because human choices are part of the cause, human choices can also be part of the solution. Both big actions (by countries and companies) and small everyday habits add up.

Big Solutions	Everyday Habits
Switch to renewable energy (solar, wind, hydro)	Turn off lights and unplug devices
Build more efficient vehicles and buildings	Walk, bike, carpool, or use the bus
Protect and plant forests (trees absorb CO ₂)	Reduce, reuse, and recycle
Reduce factory pollution	Save water and electricity at home

★ Trees Are Natural Helpers

Plants take in carbon dioxide and release oxygen. That's why protecting forests and planting trees is one of the most natural ways to remove some greenhouse gas from the air — the same plants that make our oxygen also help cool the planet.

? Quick Check 6.6

1. Name one big-scale solution and one everyday habit that reduce greenhouse gases.
2. How do trees help with greenhouse gases?

Chapter 6 Wrap-Up

★ Key Takeaways

- **Natural resources** are things from Earth that people use; they're either renewable or nonrenewable.
- **Renewable resources** (solar, wind, water, geothermal, biomass, tidal) refill naturally and are sustainable.
- **Nonrenewable resources** (coal, oil, natural gas, uranium) are limited and take millions of years to form.
- **Conservation** protects water, soil, and air; the Three R's help us use fewer resources.
- **Global temperatures have risen** over the past century, closely tied to greenhouse gases from burning fossil fuels — and our choices can help.